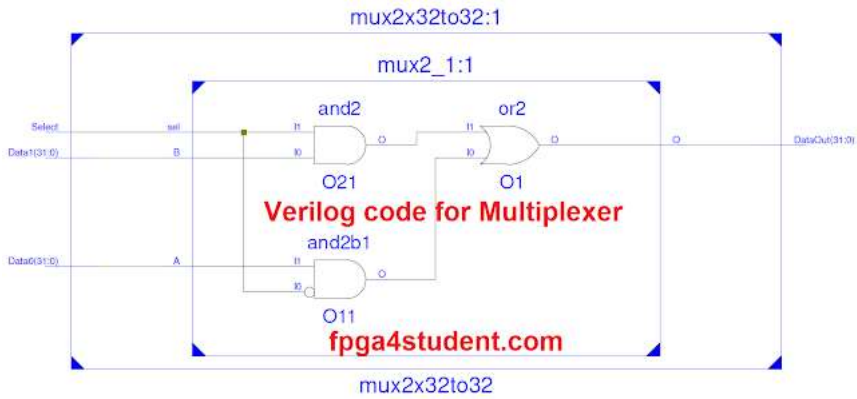
Verilog code for Multiplexers

Multiplexers are one of the main combinational logic components in digital circuits. Multiplexers are used for selecting one of many different digital inputs and forwarding to the output based on the controlling signals.

In this Verilog project, Verilog code for multiplexers such as 2-to-1 multiplexer, 2x5-to-5 multiplexer and 2x32-to-32 multiplexer are presented.



Verilog code for Multiplexers:

```
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// Verilog project: Verilog code for Multiplexer
// Verilog code for 2x32-to-32 Multiplexer
module mux2x32to32( DataOut,Data0, Data1, Select);
output [31:0] DataOut; // Data Out
input [31:0] Data0, Data1; // Data In 1 and 2
input Select;
// if Select = 0, DataOut = Data0
// otherwise, DataOut = Data1

mux2_1 mux0(DataOut[0],Data0[0],Data1[0],Select);
mux2_1 mux1(DataOut[1],Data0[1],Data1[1],Select);
mux2_1 mux2(DataOut[2],Data0[2],Data1[2],Select);
mux2_1 mux3(DataOut[3],Data0[3],Data1[3],Select);
mux2_1 mux4(DataOut[4],Data0[4],Data1[4],Select);
mux2_1 mux5(DataOut[5],Data0[5],Data1[5],Select);
mux2_1 mux6(DataOut[6],Data0[6],Data1[6],Select);
mux2_1 mux7(DataOut[7],Data0[7],Data1[7],Select);
mux2_1 mux8(DataOut[8],Data0[8],Data1[8],Select);
mux2_1 mux9(DataOut[9],Data0[9],Data1[9],Select);
mux2_1 mux10(DataOut[10],Data0[10],Data1[10],Select);
mux2_1 mux11(DataOut[11],Data0[11],Data1[11],Select);
mux2_1 mux12(DataOut[12],Data0[12],Data1[12],Select);
mux2_1 mux13(DataOut[13],Data0[13],Data1[13],Select);
mux2_1 mux14(DataOut[14],Data0[14],Data1[14],Select);
mux2_1 mux15(DataOut[15],Data0[15],Data1[15],Select);
mux2_1 mux16(DataOut[16],Data0[16],Data1[16],Select);
mux2_1 mux17(DataOut[17],Data0[17],Data1[17],Select);
mux2_1 mux18(DataOut[18],Data0[18],Data1[18],Select);
mux2_1 mux19(DataOut[19],Data0[19],Data1[19],Select);
mux2_1 mux20(DataOut[20],Data0[20],Data1[20],Select);
mux2_1 mux21(DataOut[21],Data0[21],Data1[21],Select);
mux2_1 mux22(DataOut[22],Data0[22],Data1[22],Select);
mux2_1 mux23(DataOut[23],Data0[23],Data1[23],Select);
mux2_1 mux24(DataOut[24],Data0[24],Data1[24],Select);
mux2_1 mux25(DataOut[25],Data0[25],Data1[25],Select);
mux2_1 mux26(DataOut[26],Data0[26],Data1[26],Select);
```

```
mux2_1 mux27(DataOut[27],Data0[27],Data1[27],Select);
mux2_1 mux28(DataOut[28],Data0[28],Data1[28],Select);
mux2_1 mux29(DataOut[29],Data0[29],Data1[29],Select);
mux2_1 mux30(DataOut[30],Data0[30],Data1[30],Select);
mux2_1 mux31(DataOut[31],Data0[31],Data1[31],Select);

endmodule
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// Verilog project: Verilog code for Multiplexer
// Verilog code for 2x5-to-5 Multiplexer
module mux2x5to5( AddrOut,Addr0, Addr1, Select);
output [4:0] AddrOut; // Address Out
input [4:0] Addr0, Addr1; // Address In 1 and 2
input Select;
mux2_1 mux0(AddrOut[0],Addr0[0],Addr1[0],Select);
mux2_1 mux1(AddrOut[1],Addr0[1],Addr1[1],Select);
mux2_1 mux2(AddrOut[2],Addr0[2],Addr1[2],Select);
mux2_1 mux3(AddrOut[3],Addr0[3],Addr1[3],Select);
mux2_1 mux4(AddrOut[4],Addr0[4],Addr1[4],Select);
endmodule
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// Verilog project: Verilog code for Multiplexer
// Verilog code for 2-to-1 Multiplexer
module mux2_1(O,A,B,sel);
// if sel = 0, O = A
// if sel = 1, O =B
output O;
input A,B,sel;
not #(50) not1(nsel,sel);
and #(50) and1(O1,A,nsel);
and #(50) and2(O2,B,sel);
or #(50) or2(O,O1,O2);
endmodule
```

The Multiplexer is implemented and successfully verified in the Verilog project of 32-bit MIPS processor here. The Multiplexer is used in register file and the top level of the pipelined MIPS processor.

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26. 32-bit 5-stage Pipelined MIPS Processor in Verilog (Part-2)
27. 32-bit 5-stage Pipelined MIPS Processor in Verilog (Part-3)
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